

Mark schemes

Q1.

(a) $a = \frac{dv}{dt} = 12t + 4$

M1 A1

2

(b) Using $F = ma$,
Force = $3 \times (12t + 4)$

M1

When $t = 4$, force = $3 (12 \times 4 + 4)$
Force = 156 N

A1

2

[4]

Q2.

(a) $a = \frac{dv}{dt} = 12t + 4$

M1A1

2

(b) Using $F = ma$,
Force = $3 \times (12t + 4)$

M1

When $t = 4$, force = $3 (12 \times 4 + 4)$
Force = 156 N

A1

2

(c) $r = 2t^3 + 2t^2 - 7t + c$

M1A1

When $t = 0$, $r = 5$, $\therefore c = 5$

M1

$\therefore r = 2t^3 + 2t^2 - 7t + 5$

SC3 if no '+c' seen

A1

4

[8]

Q3.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1

2

(b) (i) $\text{Speed} = \sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1

2

(ii) $\text{Maximum speed} = \sqrt{1.2^2 + 0.9^2}$

Use of $\sin t = 1$

m1

$= 1.5$

A1

2

[6]

Q4.

(a) (i) $a = 2 + 12e^{-t}$

Differentiating, with at least one term correct.

Correct velocity

M1A1

2

(ii) $2 < a \leq 14$

For 2, For 14

B1, B1

Correct inequalities

B1

3

(b) $s = t^2 + 12e^{-t} + c$

Integrating, with at least one term correct.

M1

Correct expression with or without c

A1

$$s = 0, t = 0 \Rightarrow c = -12$$

Finding c

dM1

$$s = t^2 + 12e^{-t} - 12$$

Correct final expression

A1

4

[9]

Q5.

(a) $\mathbf{a} = 8 \cos t \mathbf{i} - 4 \sin t \mathbf{j}$

M1 differentiation

M1A1

$$\mathbf{F} = 2 \cos t \mathbf{i} - \sin t \mathbf{j}$$

A1

3

(b) (i) $|\mathbf{F}| = \sqrt{4 \cos^2 t + \sin^2 t}$

Magnitude

M1

$$= \sqrt{3 \cos^2 t + 1}$$

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CAO

A1

3

(ii) $|\mathbf{F}| \leq 4$

B1 each value (at end of range)

B1B1

2

[8]

Q6.

(a) (i) $\mathbf{v} = 2 \cos 2t \mathbf{i} + 6 \mathbf{j}$

M1 differentiation (6t)

M1A1

2

(ii) $|\mathbf{v}| = \sqrt{4 \cos^2 2t + 36}$
Sum of squares, for v or v²

M1

ft trig term v

A1F

CAO

A1

3

(iii) $\cos^2 2t = 0$ or $\cos 2t = 0$

M1

$t = \frac{\pi}{4}$
radians

A1

2

(b) (i) $\mathbf{a} = -4 \sin 2t \mathbf{i}$
Differentiation attempt

M1

$F = 0.25a$
Used

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M1

$F = -\sin 2t \mathbf{i}$
ft v, see vector

A1F

3

(ii) Direction is $\pm \mathbf{i}$

B1

$|\sin 2t| \leq 1$

B1

2

[12]

Q7.

(a) $\frac{dv}{dt} = 6t - 5$

M1 A1

2

(b) $6t - 5 = 0$

$$t = \frac{5}{6}$$

A1F

1

[3]

Q8.

(a) $\mathbf{v} = -4e^{-t} \mathbf{i} + (6 - 3e^{-t}) \mathbf{j}$

Differentiating position vector

M1

$t = 0$ A1

Correct velocity

$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j}$

ag Substituting $t = 0$ to obtain initial velocity

A1

3

(b) $\mathbf{a} = 4e^{-t} \mathbf{i} + 3e^{-t} \mathbf{j}$

Differentiating velocity

M1

Correct acceleration

A1

2

(c) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Finding acceleration when $t = 0$

M1

$$a = \sqrt{4^2 + 3^2} = 5$$

Correct magnitude

A1

2

(d) $\mathbf{v} \rightarrow 0\mathbf{i} + 6\mathbf{j}$

For i component

B1

For j component

B1

2

[9]

Q9.

(a) (i) $s(10) = 25 - 100 + 150 = 75$

Correct distance

B1

1

(ii) $v = \frac{t^3}{100} - \frac{3t^2}{10} + 3t$

Differentiating s

Correct derivative

M1 A1

$v(10) = 10 - 30 + 30 = 10$

Correct v

A1

3

(iii) $a = \frac{3t^2}{100} - \frac{3t}{5} + 3$

Differentiating v

Correct derivative

M1 A1

$a(10) = 3 - 6 + 3 = 0$

Correct a

A1

3

(b) $h = 10$

Value of h

B1

$75 = 100 - k$

Substituting s = 75 and t = 10

M1

$$k = 25$$

Correct k

A1

3

[10]

Q10.

(a) $\mathbf{r} = 5\mathbf{j}$

Correct vector

B1

1

(b) $4t - 0.01t^2 = 0$

Equation based on i component

M1

Correct equation

A1

$$t = 0 \text{ or } t = 400$$

Solving the quadratic

M1

$$t = 400$$

Selecting $t = 400$

A1

4

(c) $\mathbf{v} = (4 - 0.02t)\mathbf{i} + (-3 - 0.08t)\mathbf{j}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

Correct velocity vector

A1

$$4 - 0.02t = 3 + 0.08t$$

Equation using both components

M1

Correct equation

A1

$$t = 10$$

Correct t

A1

5

[10]

Q11.

(a) $v = 4 - 2e^{-t}$

Differentiating

M1

Correct velocity

A1

2

(b) $a = 2e^{-t}$

Differentiating

M1

Correct acceleration

A1

2

(c) Acceleration decreases, approaching a value of zero

Decreasing

B1

Approaching zero

B1

2

[6]

Q12.

(a) $\mathbf{v} = -2\sin\left(\frac{t}{20}\right)\mathbf{i} + 4\cos\left(\frac{t}{20}\right)\mathbf{j}$

Differentiating

i component correct

j component correct

M1 A1 A1

3

(b) $\mathbf{v} = 4\mathbf{j}$

Correct velocity at $t = 0$

B1

Travelling north

Travelling north with $\mathbf{v} = nj$ where $n > 0$

B1

2

(c) $2 \sin\left(\frac{t}{20}\right) = 0$

Equation for t based on 0i

M1

$$\frac{t}{20} = \pi$$

$$t = 20\pi$$

Correct solution

A1

2

[7]



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